

Analysts can determine what is considered to be an acceptable level of cross factor correlation for their particular needs and determine the data sample set within these criteria through a random sampling process.

Heteroscedasticity. Analysis of the complete model above reveals potential heteroscedasticity of the error term. Each stock in the sample has different volatilities and POV rates (resulting in different trading times) and a different distribution of the error term. Kissell (2006) provides techniques to correct for heteroscedasticity in this model. One important note, however, is that after grouping the data into bins there is not much difference between the parameter estimation results without correcting for heteroscedasticity. We highly recommend analysts perform both analyses to understand the dynamics of this model before deciding if the heteroscedasticity step can be ignored.

Grouping Data. Prior to performing our non-linear regression we grouped our data into buckets in order to average away noise and to ensure a balanced data set. Data was bucketed into categories based on size, volatility, and POV rate according to the following criteria:

Size = 0.5%, 1%, 2%, ..., 30%

Volatility = 10%, 20%, ..., 80%

POV Rate = 1%, 5%, 10%, ..., 65%

If we use too fine increments excessive groupings surface and we have found that a substantially large data grouping does not always uncover a statistical relationship between cost and our set of explanatory factors.

Next we averaged costs for each category above. We required at least 25 observations in order for the bucket to be included in the regression analysis. We required at least 25 data points in order to average away noise and determine the most likely cost given the category of size, volatility, and POV rate.

Sensitivity Analysis

We discussed briefly that we need to ensure a solution with a feasible set of parameter values. In other words, we are setting constraints on the model parameters. These feasible values are:

$$100 \leq a_1 \leq 1000$$

$$0.10 \leq a_2 \leq 1.0$$

$$0.10 \leq a_3 \leq 1.0$$

$$0.10 \leq a_4 \leq 1.0$$

$$0.70 \leq b_1 \leq 1$$